INSTALLING MESSAGE PASSING INTERFACE (MPI) - CLUSTER

MPI

Command: sudo su
Command: sudo apt install mpich openmpi-bin libopenmpi-dev -y
TESTING THE INSTALLATION
Command: mpicc.openmpi --version
Command: mpicc --showme:version
Command: nproc
Command: mpirun --version
Command: mpirun hostname
Command: mpirun uname
Command: mpirun -np 4 hostname

MPI Python

Command: sudo apt-get install python3-mpi4py -y
Command: sudo apt install python3-pip -y
Command: pip3 install mpi4py
Command: pip3 install mpi4py --upgrade
Command: pip3 uninstall mpi4py

TESTING THE INSTALLATION
Command: mpiexec -n 4 python3 -m mpi4py.bench helloworld
Command: python3
>>> import mpi4py
>>> exit()

SSH

Command: sudo apt-get install openssh-server -y
TESTING THE INSTALLATION
Command: sudo systemctl status ssh
Command: sudo ufw allow ssh

Command: ip addr show
Command: ssh user@ip

RESETANDO LISTA
Command: ssh-keygen -R [hostname]
Command: ssh-keygen -R [ip]

TURN OFF THE NODES
Command: sudo shutdown now

Configure SSH for MPI

Command: ssh server@10.0.0.2
Command: ssh node1@10.0.0.3
Command: ssh node2@10.0.0.4
== SERVER ==
Command: touch machines
Command: nano machines
server@10.0.0.2 slots=12
node1@10.0.0.3 slots=12
node2@10.0.0.4 slots=12
Command: cat machines
Command: cd ~
Command: ssh-keygen
Command: cd .ssh
Command: ls
Command: cp id_rsa.pub server
Command: ls
Command: cd ~
Command: sudo mkdir /home/Python
== NODE 1 ==
Command: cd ~
Command: ssh-keygen
Command: cd .ssh
Command: ls
Command: cp id_rsa.pub node1
Command: scp server@10.0.0.2:/home/server/.ssh/server .
Command: yes
Command: cat server>>authorized_keys
Command: cd ~
Command: sudo mkdir /home/Python
== NODE 2 ==
Command: cd ~
Command: ssh-keygen
Command: cd .ssh
Command: ls
Command: cp id_rsa.pub node2
Command: scp server@10.0.0.2:/home/server/.ssh/server .
Command: cat server>>authorized_keys
Command: cd ~
Command: sudo mkdir /home/Python
== SERVER ==
Command: scp node1@10.0.0.3:/home/node1/.ssh/node1 .
Command: scp node2@10.0.0.4:/home/node2/.ssh/node2 .
Command: cat node1>>authorized_keys
Command: cat node2>>authorized_keys
== TEST ==
Command: mpirun -hostfile machines -np 36 hostname

TRANSFERRING FILES TO NODES

== SERVER ==

Command: scp /home/eder/Downloads/{filename.py}node1@10.0.0.3:/home/node1/Downloads

Command: scp /home/eder/Downloads/{filename.py}node2@10.0.0.4:/home/node2/Downloads

Command: sudo cp /home/server/Downloads/{filename.py} /home/Python/{filename.py}

== NODE 1 ==

Command: ssh node1@10.0.0.3

Command: sudo cp /home/node1/Downloads/{filename.py} /home/Python/{filename.py}

== NODE 2 ==

Command: ssh node2@10.0.0.4

Command: sudo cp /home/node2/Downloads/{filename.py} /home/Python/{filename.py}

== SERVER ==

Command: mpirun -hostfile machines -np 36 python3 /home/Python/{filename.py}

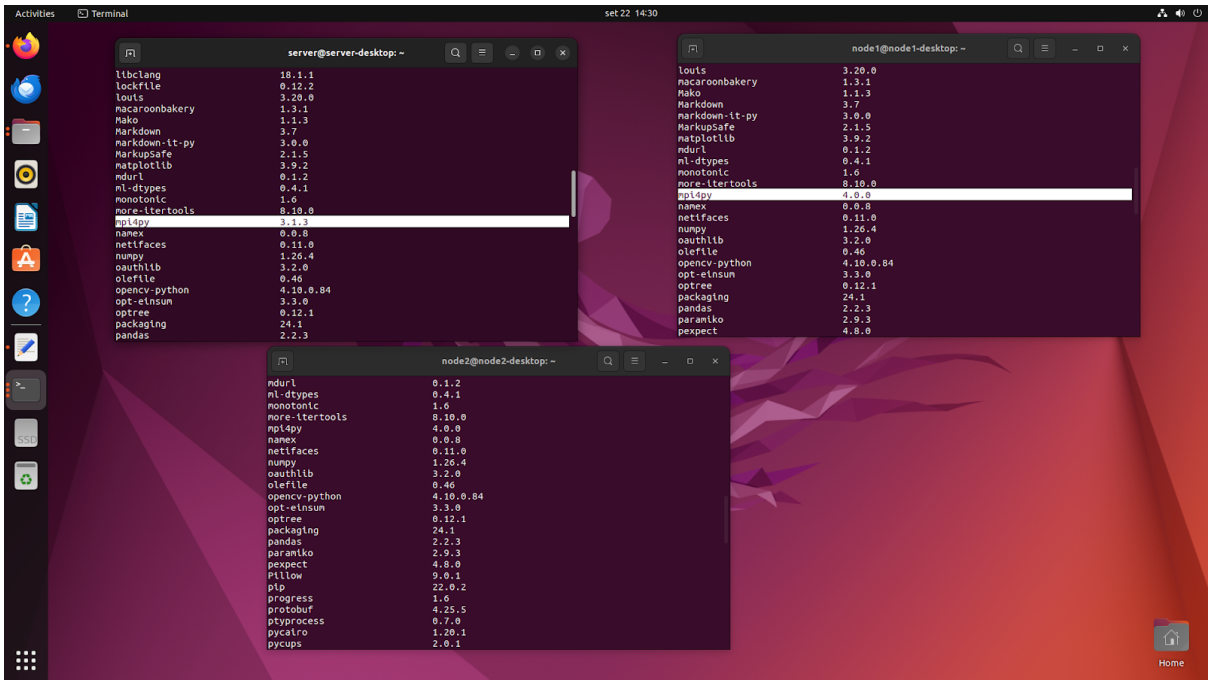
CONFIGURING ROUTER

http://10.0.0.1

admin

3twgnids

POSSIBLE ERRORS DUE TO SOFTWARE VERSIONS



PYTHON INSTALL

Command: pip list

GENERIC

```
pip3 install seaborn
pip3 install opencv-python
pip3 install tensorflow
pip3 install tqdm
pip3 install scipy
```

GRAPHIC INTERFACE

```
pip3 install matplotlib
pip3 install latex
pip3 install numpy
pip3 install numpy-linalg
```

DATA ACQUISITION

```
pip3 install pyModbusTCP
pip3 install pyModbusTCP==v0.1.10
pip3 install pyserialpip3 install progress
```